

## LAB PRACTICE – CO4 AT2

### JAVA PROGRAMMING

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#### 1. JAVA APPLET LIFECYCLE

##### Java Applet Lifecycle Program

```
import java.applet.Applet;
import java.awt.Graphics;

public class LifeCycleApplet extends Applet {

    public void init() {
        System.out.println("init() called");
    }

    public void start() {
        System.out.println("start() called");
    }

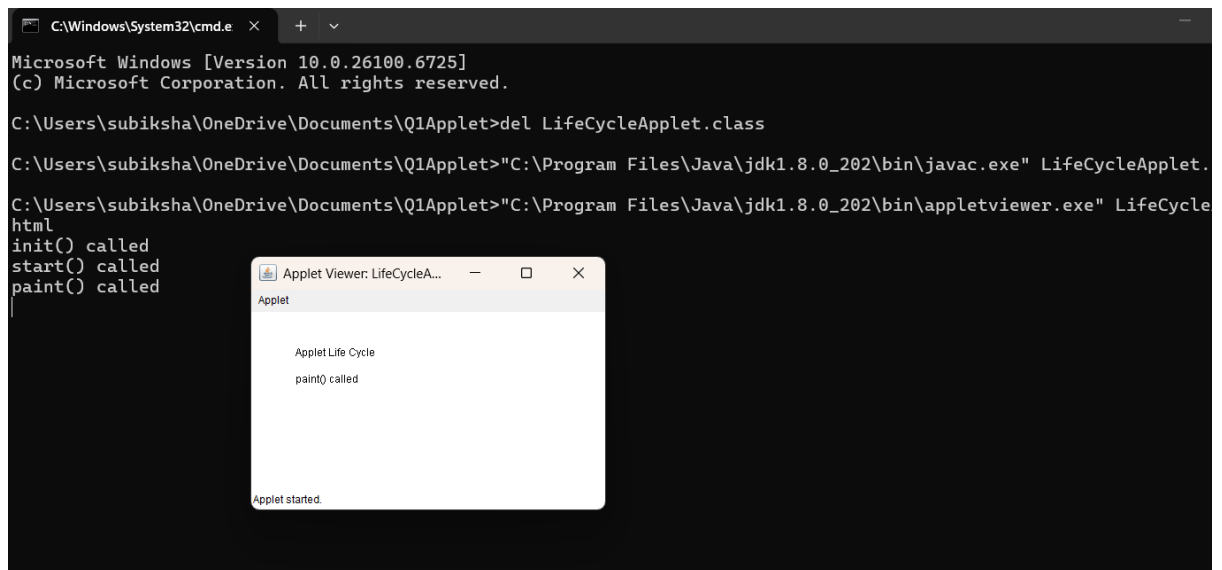
    public void paint(Graphics g) {
        g.drawString("Applet Life Cycle", 50, 50);
        g.drawString("paint() called", 50, 80);
        System.out.println("paint() called");
    }

    public void stop() {
        System.out.println("stop() called");
    }

    public void destroy() {
        System.out.println("destroy() called");
    }
}
```

## HTML File

```
<html>
<body>
<applet code="AppletLifeCycle.class" width="400" height="300">
</applet>
</body>
</html>
```



## 2. Generating Graphic Images Using Applet GraphicsApplet.java

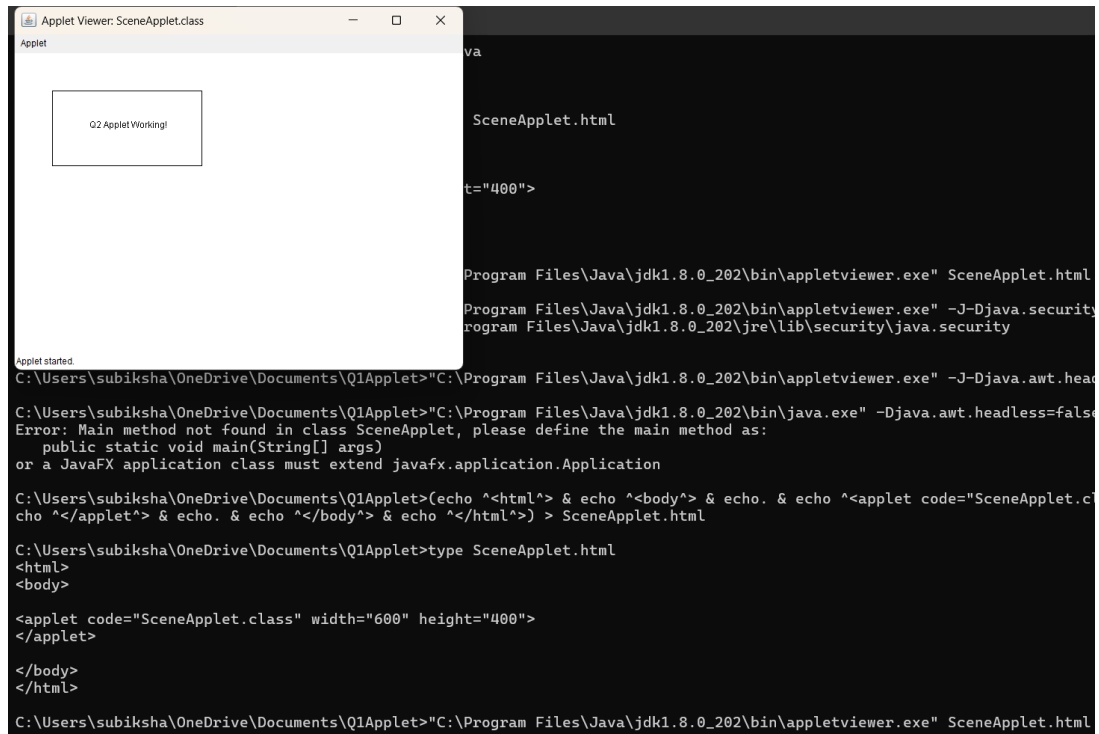
```
import java.applet.Applet;
import java.awt.Graphics;

public class SceneApplet extends Applet {

    public void paint(Graphics g) {
        g.drawString("Q2 Applet Working!", 100, 100);
        g.drawRect(50, 50, 200, 100);
    }
}
```

## HTML

```
<html>
<body>
<applet code="SceneApplet.class" width="600" height="400">
</applet>
</body>
</html>
```



### 3. Java Applet Calculator Using Parameters

#### CalculatorApplet.java

```
import java.applet.Applet;

import java.awt.Graphics;

public class CalculatorApplet extends Applet
{
    int num1, num2, sum, sub, mul, div;

    public void init()
    {
        num1 = Integer.parseInt(getParameter("num1"));
        num2 = Integer.parseInt(getParameter("num2"));

        sum = num1 + num2;

        sub = num1 - num2;

        mul = num1 * num2;

        if(num2 != 0)
            div = num1 / num2;
    }

    public void paint(Graphics g)
    {
        g.drawString("Calculator Using Applet Parameters", 50, 50);

        g.drawString("First Number = " + num1, 50, 80);

        g.drawString("Second Number = " + num2, 50, 110);

        g.drawString("Addition = " + sum, 50, 150);
```

```
g.drawString("Subtraction = " + sub, 50, 180);

g.drawString("Multiplication = " + mul, 50, 210);

if(num2 != 0)

    g.drawString("Division = " + div, 50, 240);

else

    g.drawString("Division by Zero Not Possible", 50, 240);

}

}
```

### **CalculatorApplet.html**

```
<html>

<body>

<applet code="CalculatorApplet.class"

    width="500"

    height="300">

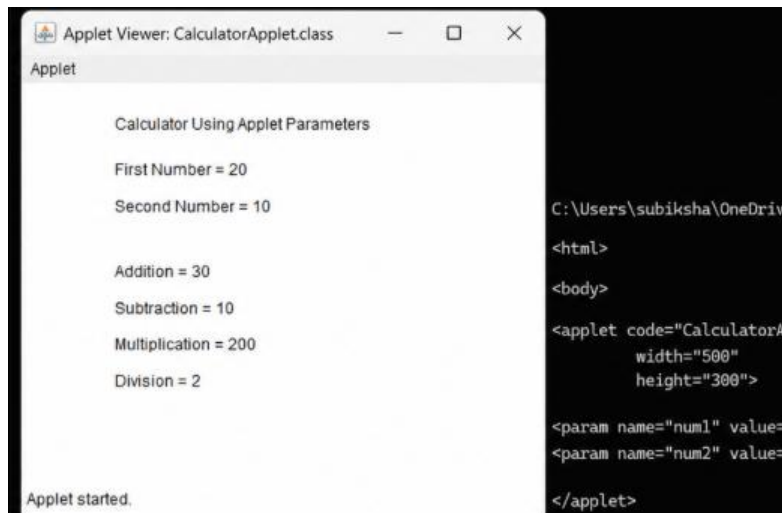
<param name="num1" value="20">

<param name="num2" value="10">

</applet>

</body>

</html>
```



#### 4. Student Profile Using AWT Components

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class StudentProfile extends Frame implements ActionListener {
```

```
    Label nameLabel, regLabel, genderLabel;
```

```
    Label deptLabel, yearLabel, skillsLabel;
```

```
    TextField nameField;
```

```
    TextField regField;
```

```
    CheckboxGroup genderGroup;
```

```
    Checkbox male, female;
```

```
    Choice department;
```

```
    Choice year;
```

```
    Checkbox java, python, cpp;
```

```
    TextArea reportArea;
```

```
    Button generateButton;
```

```
    Button clearButton;
```

```
    List skillList;
```

```
public StudentProfile() {

    setTitle("Student Profile");

    setSize(600, 600);

    setLayout(new FlowLayout());

    nameLabel = new Label("Name:");

    nameField = new TextField(30);

    regLabel = new Label("Register Number:");

    regField = new TextField(20);

    genderLabel = new Label("Gender:");

    genderGroup = new CheckboxGroup();

    male = new Checkbox("Male", genderGroup, false);

    female = new Checkbox("Female", genderGroup, false);

    deptLabel = new Label("Department:");

    department = new Choice();

    department.add("AI & DS");

    department.add("CSE");

    department.add("ECE");

    department.add("IT");

    department.add("EEE");


    // Year

    yearLabel = new Label("Year:");


    year = new Choice();

    year.add("1st Year");

    year.add("2nd Year");
```

```
year.add("3rd Year");

year.add("4th Year");


// Skills

skillsLabel = new Label("Skills:");


java = new Checkbox("Java");

python = new Checkbox("Python");

cpp = new Checkbox("C++");


// List

skillList = new List(4);

skillList.add("Java");

skillList.add("Python");

skillList.add("C++");

skillList.add("SQL");


// Report area

reportArea = new TextArea(10, 45);


// Buttons

generateButton = new Button("Generate Report");

clearButton = new Button("Clear");


generateButton.addActionListener(this);

clearButton.addActionListener(this);
```



```
// Add components

add(nameLabel);

add(nameField);

add(regLabel);

add(regField);

add(genderLabel);

add(male);

add(female);

add(deptLabel);

add(department);

add(yearLabel);

add(year);

add(skillsLabel);

add(java);

add(python);

add(cpp);

add(new Label("Programming Languages:"));

add(skillList);


add(generateButton);

add(clearButton);

add(reportArea);

addWindowListener(new WindowAdapter() {

    public void windowClosing(WindowEvent e) {

        dispose();

    }

});
```

```

    }

});

setVisible(true);
}

public void actionPerformed(ActionEvent e) {

    if (e.getSource() == generateButton) {

        String gender = "";

        if (male.getState())

            gender = "Male";

        else if (female.getState())

            gender = "Female";

        String skills = "";

        if (java.getState())

            skills += "Java ";

        if (python.getState())

            skills += "Python ";

        if (cpp.getState())

            skills += "C++ ";

        reportArea.setText(

            "STUDENT PROFILE\n\n" +

            "Name: " + nameField.getText() + "\n" +

            "Register Number: " + regField.getText() + "\n" +

            "Gender: " + gender + "\n" +

            "Department: " + department.getSelectedItemAt() + "\n" +

            "Year: " + year.getSelectedItemAt() + "\n" +

            "Skills: " + skills + "\n" +

```

```

        "Selected List Item: " +

        skillList.getSelectedItemAt(

    );

}

else if (e.getSource() == clearButton) {

    nameField.setText("");

    regField.setText("");

    reportArea.setText("");

    male.setState(false);

    female.setState(false);

    java.setState(false);

    python.setState(false);

    cpp.setState(false);

}

}

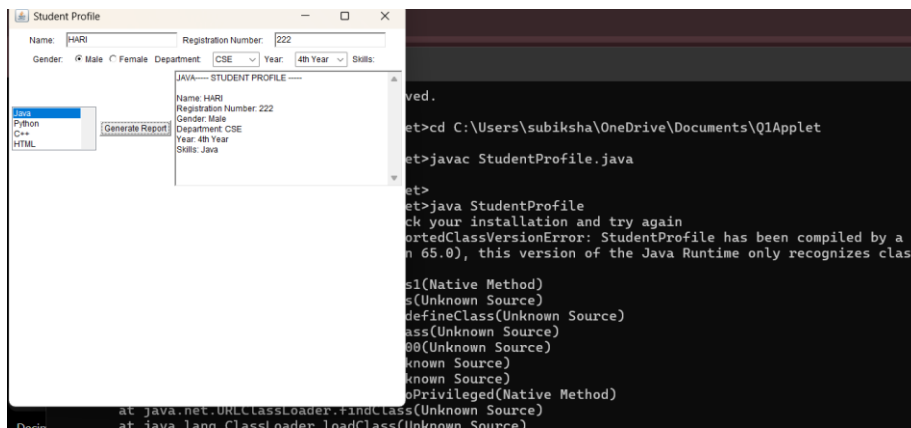
public static void main(String[] args) {

    new StudentProfile();

}

}

```



## 5. GridLayout with Event Handling

### CODE

```
import java.awt.*;

import java.awt.event.*;

public class GridCalculator extends Frame implements ActionListener {

    TextField display;

    double num1 = 0;

    double num2 = 0;

    String operator = "";

    public GridCalculator() {

        setTitle("GridLayout Calculator");

        setSize(400, 500);

        setLayout(new BorderLayout());

        display = new TextField();

        display.setEditable(false);

        add(display, BorderLayout.NORTH);

        Panel panel = new Panel();

        panel.setLayout(new GridLayout(4, 4, 5, 5));

        String[] buttons = {

            "7", "8", "9", "+",

            "4", "5", "6", "-",

            "1", "2", "3", "*",

            "0", "=", "C", "/"

        };

        for (String text : buttons) {

            Button button = new Button(text);

            button.addActionListener(this);
```

```

        panel.add(button);
    }
    add(panel, BorderLayout.CENTER);
    addWindowListener(new WindowAdapter() {
        public void windowClosing(WindowEvent e) {
            dispose();
        }
    });
    setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    String value = e.getActionCommand();

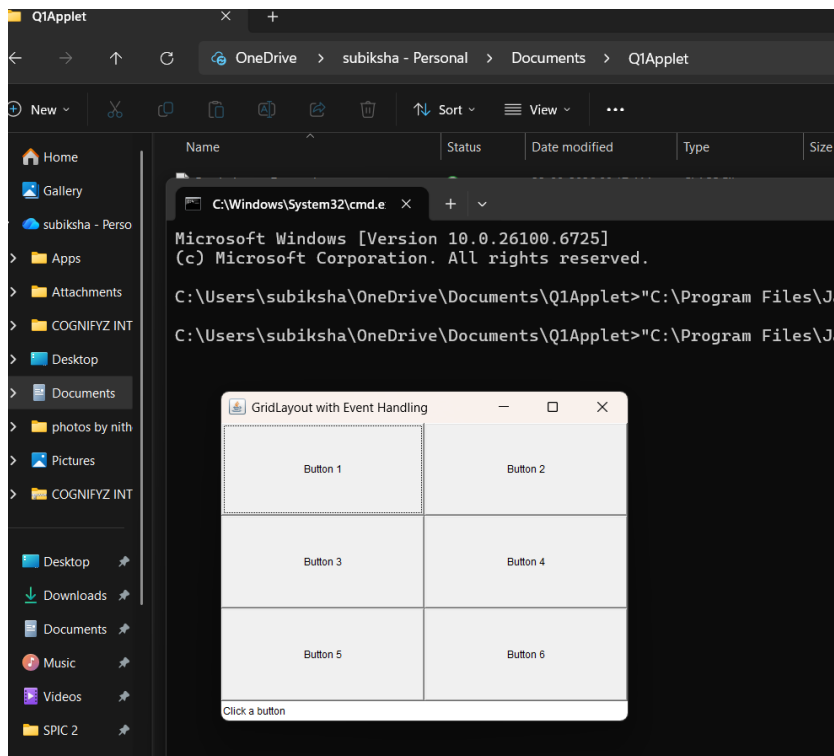
    if (value.equals("C")) {
        display.setText("");
        num1 = 0;
        num2 = 0;
        operator = "";
    }
    else if (value.equals("+") ||
            value.equals("-") ||
            value.equals("*") ||
            value.equals("/")) {
        num1 = Double.parseDouble(display.getText());
        operator = value;
        display.setText("");
    }
    else if (value.equals("=")) {
        num2 = Double.parseDouble(display.getText());

```

```

double result = 0;
if (operator.equals("+"))
    result = num1 + num2;
else if (operator.equals("-"))
    result = num1 - num2;
else if (operator.equals("*"))
    result = num1 * num2;
else if (operator.equals("/")) {
    if (num2 != 0)
        result = num1 / num2;
    else {
        display.setText("Cannot divide by zero");
        return;
    }
}
display.setText(String.valueOf(result));
}
else {
    display.setText(display.getText() + value);
}
}
public static void main(String[] args) {
    new GridCalculator();
}
}

```



6.

**CODE :**

```
import java.awt.*;
import java.awt.event.*;

public class StudentRegistrationForm extends Frame implements ActionListener {

    static final Color DEEP_NAVY = new Color(13, 22, 40);
    static final Color NAVY_LIGHT = new Color(22, 36, 64);
    static final Color GOLD = new Color(212, 175, 55);
    static final Color GOLD_SOFT = new Color(240, 217, 150);
    static final Color CREAM = new Color(250, 246, 236);
    static final Color MAROON = new Color(90, 24, 32);
    static final Color EMERALD = new Color(20, 92, 71);

    Label nameLabel, rollLabel, deptLabel, emailLabel, phoneLabel, genderLabel, courseLabel;
    TextField nameField, rollField, emailField, phoneField;

    Choice deptChoice, courseChoice;

    Checkbox maleBox, femaleBox;

    CheckboxGroup genderGroup;
```

```

Button submitButton, clearButton, exitButton;

TextArea outputArea;

GradientPanel formPanel;

StudentRegistrationForm() {

    setTitle("Student Registration | Elite Academy");

    setSize(600, 660);

    setLayout(null);

    setResizable(false);

    formPanel = new GradientPanel();

    formPanel.setLayout(null);

    formPanel.setBounds(0, 0, 600, 660);

    add(formPanel);

    Label crest = new Label("\u2726", Label.CENTER);

    crest.setFont(new Font("Serif", Font.BOLD, 26));

    crest.setForeground(GOLD);

    crest.setBounds(275, 22, 50, 30);

    formPanel.add(crest);

    Label titleLabel = new Label("ELITE ACADEMY", Label.CENTER);

    titleLabel.setFont(new Font("Serif", Font.BOLD, 30));

    titleLabel.setForeground(GOLD);

    titleLabel.setBounds(100, 52, 400, 40);

    formPanel.add(titleLabel);

    Label subtitleLabel = new Label("S T U D E N T   R E G I S T R A T I O N", Label.CENTER);

    subtitleLabel.setFont(new Font("SansSerif", Font.PLAIN, 13));

    subtitleLabel.setForeground(CREAM);

    subtitleLabel.setBounds(100, 90, 400, 22);

    formPanel.add(subtitleLabel);

    Divider div1 = new Divider();

    div1.setBounds(80, 120, 440, 2);

    formPanel.add(div1);

    int labelX = 70, fieldX = 230, fieldW = 300, rowH = 42, startY = 150;

```



```
nameLabel = luxLabel("Full Name");
nameLabel.setBounds(labelX, startY, 150, 24);
formPanel.add(nameLabel);
nameField = luxField();
nameField.setBounds(fieldX, startY - 3, fieldW, 28);
formPanel.add(nameField);
```

```
rollLabel = luxLabel("Roll Number");
rollLabel.setBounds(labelX, startY + rowH, 150, 24);
formPanel.add(rollLabel);
rollField = luxField();
rollField.setBounds(fieldX, startY + rowH - 3, fieldW, 28);
formPanel.add(rollField);
```

```
emailLabel = luxLabel("Email Address");
emailLabel.setBounds(labelX, startY + rowH * 2, 150, 24);
formPanel.add(emailLabel);
emailField = luxField();
emailField.setBounds(fieldX, startY + rowH * 2 - 3, fieldW, 28);
formPanel.add(emailField);
```

```
phoneLabel = luxLabel("Phone Number");
phoneLabel.setBounds(labelX, startY + rowH * 3, 150, 24);
formPanel.add(phoneLabel);
phoneField = luxField();
phoneField.setBounds(fieldX, startY + rowH * 3 - 3, fieldW, 28);
formPanel.add(phoneField);
```

```
deptLabel = luxLabel("Department");
deptLabel.setBounds(labelX, startY + rowH * 4, 150, 24);
formPanel.add(deptLabel);
```

```
deptChoice = new Choice();
deptChoice.add("Computer Science");
deptChoice.add("Electronics & Comm.");
deptChoice.add("Mechanical");
deptChoice.add("Civil");
deptChoice.add("Information Technology");
deptChoice.setBounds(fieldX, startY + rowH * 4 - 3, fieldW, 28);
deptChoice.setBackground(CREAM);
deptChoice.setForeground(DEEP_NAVY);
formPanel.add(deptChoice);
```

```
courseLabel = luxLabel("Course");
courseLabel.setBounds(labelX, startY + rowH * 5, 150, 24);
formPanel.add(courseLabel);
courseChoice = new Choice();
courseChoice.add("B.E / B.Tech");
courseChoice.add("M.E / M.Tech");
courseChoice.add("MBA");
courseChoice.add("MCA");
courseChoice.setBounds(fieldX, startY + rowH * 5 - 3, fieldW, 28);
courseChoice.setBackground(CREAM);
courseChoice.setForeground(DEEP_NAVY);
formPanel.add(courseChoice);
```

```
genderLabel = luxLabel("Gender");
genderLabel.setBounds(labelX, startY + rowH * 6, 150, 24);
formPanel.add(genderLabel);
```

```
genderGroup = new CheckboxGroup();
maleBox = new Checkbox("Male", genderGroup, true);
maleBox.setBounds(fieldX, startY + rowH * 6 - 2, 90, 24);
```

```
maleBox.setForeground(CREAM);
formPanel.add(maleBox);

femaleBox = new Checkbox("Female", genderGroup, false);
femaleBox.setBounds(fieldX + 110, startY + rowH * 6 - 2, 100, 24);
femaleBox.setForeground(CREAM);
formPanel.add(femaleBox);

Divider div2 = new Divider();
div2.setBounds(80, startY + rowH * 7 + 4, 440, 2);
formPanel.add(div2);

int btnY = startY + rowH * 7 + 22;

submitButton = luxButton("SUBMIT", EMERALD, GOLD_SOFT);
submitButton.setBounds(80, btnY, 130, 38);
submitButton.addActionListener(this);
formPanel.add(submitButton);

clearButton = luxButton("CLEAR", MAROON, GOLD_SOFT);
clearButton.setBounds(235, btnY, 130, 38);
clearButton.addActionListener(this);
formPanel.add(clearButton);

exitButton = luxButton("EXIT", NAVY_LIGHT, GOLD_SOFT);
exitButton.setBounds(390, btnY, 130, 38);
exitButton.addActionListener(this);
formPanel.add(exitButton);

outputArea = new TextArea("", 0, 0, TextArea.SCROLLBARS_NONE);
outputArea.setEditable(false);
outputArea.setBounds(80, btnY + 55, 440, 80);
outputArea.setBackground(new Color(255, 253, 245));
outputArea.setForeground(DEEP_NAVY);
outputArea.setFont(new Font("SansSerif", Font.PLAIN, 12));
formPanel.add(outputArea);
```

```
Label footer = new Label("\u00A9 Elite Academy \u2022 Office of Admissions",
Label.CENTER);
```

```
    footer.setFont(new Font("SansSerif", Font.ITALIC, 11));
```

```
    footer.setForeground(GOLD_SOFT);
```

```
    footer.setBounds(100, btnY + 145, 400, 20);
```

```
    formPanel.add(footer);
```

```
addWindowListener(new WindowAdapter() {
```

```
    public void windowClosing(WindowEvent e) {
```

```
        dispose();
```

```
        System.exit(0);
```

```
    }
```

```
});
```

```
setVisible(true);
```

```
}
```

```
private Label luxLabel(String text) {
```

```
    Label l = new Label(text);
```

```
    l.setFont(new Font("SansSerif", Font.BOLD, 13));
```

```
    l.setForeground(GOLD_SOFT);
```

```
    return l;
```

```
}
```

```
private TextField luxField() {
```

```
    TextField f = new TextField();
```

```
    f.setBackground(CREAM);
```

```
    f.setForeground(DEEP_NAVY);
```

```
    f.setFont(new Font("SansSerif", Font.PLAIN, 13));
```

```
    return f;
```

```
}
```

```
private Button luxButton(String text, Color bg, Color fg) {
```

```

        Button b = new Button(text);
        b.setBackground(bg);
        b.setForeground(fg);
        b.setFont(new Font("SansSerif", Font.BOLD, 13));
        return b;
    }

```

```

public void actionPerformed(ActionEvent e) {
    String command = e.getActionCommand();
    if (command.equals("SUBMIT")) {
        String name = nameField.getText();
        String roll = rollField.getText();
        String email = emailField.getText();
        String phone = phoneField.getText();
        String dept = deptChoice.getSelectedItem();
        String course = courseChoice.getSelectedItem();
        String gender = genderGroup.getSelectedCheckbox().getLabel();
        if (name.isEmpty() || roll.isEmpty()) {
            outputArea.setText("\u26A0 Please fill in Name and Roll Number.");
            return;
        }
    }
}

```

```

        outputArea.setText(
            "\u2726 Registration Confirmed \u2726\n" +
            "Name: " + name + " | Roll No: " + roll + " | Gender: " + gender + "\n" +
            "Email: " + email + " | Phone: " + phone + "\n" +
            "Department: " + dept + " | Course: " + course
        );
    }
    else if (command.equals("CLEAR")) {
        nameField.setText("");
    }
}

```

```

rollField.setText("");
emailField.setText("");
phoneField.setText("");
deptChoice.select(0);
courseChoice.select(0);
maleBox.setState(true);
outputArea.setText("");
}
else if (command.equals("EXIT")) {
    dispose();
    System.exit(0);
}
}

static class GradientPanel extends Panel {
    public void paint(Graphics g) {
        Graphics2D g2 = (Graphics2D) g;
        int w = getWidth(), h = getHeight();

        GradientPaint gp = new GradientPaint(0, 0, DEEP_NAVY, w, h, NAVY_LIGHT);
        g2.setPaint(gp);
        g2.fillRect(0, 0, w, h);
        g2.setColor(GOLD);
        g2.setStroke(new BasicStroke(3));
        g2.drawRect(10, 10, w - 21, h - 21);
        g2.setStroke(new BasicStroke(1));
        g2.drawRect(16, 16, w - 33, h - 33);
        super.paint(g);
    }
}

static class Divider extends Canvas {
    public void paint(Graphics g) {

```

```

        g.setColor(GOLD);

        g.fillRect(0, 0, getWidth(), getHeight());

    }

}

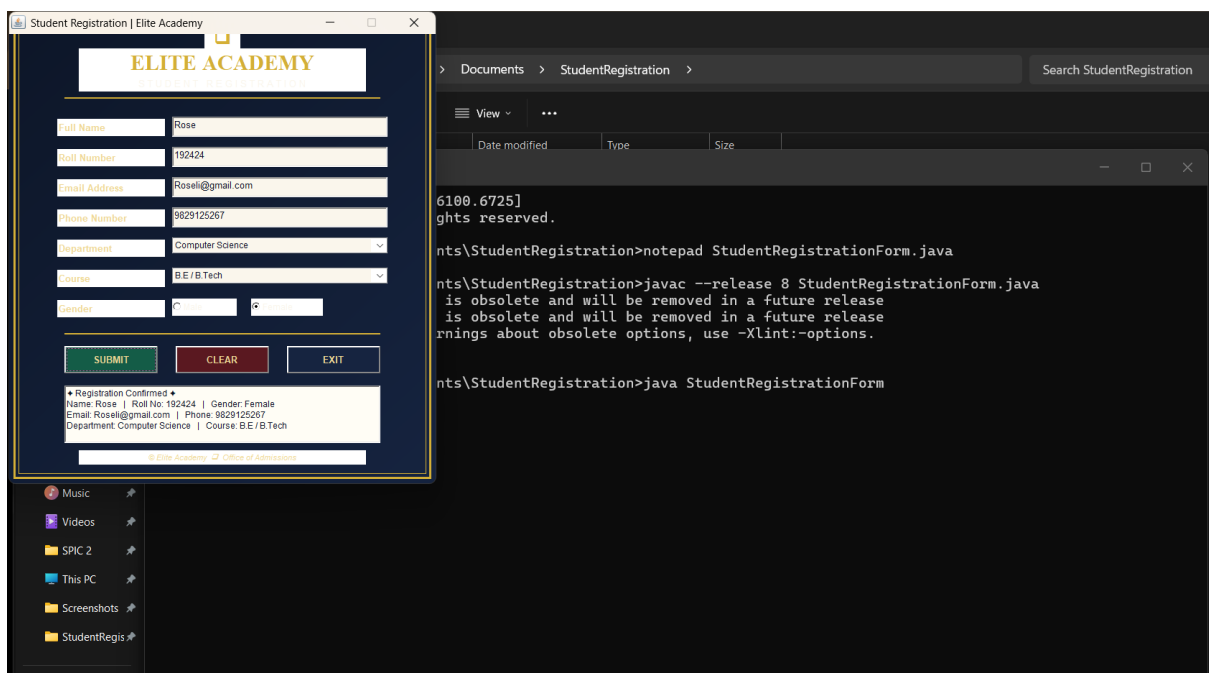
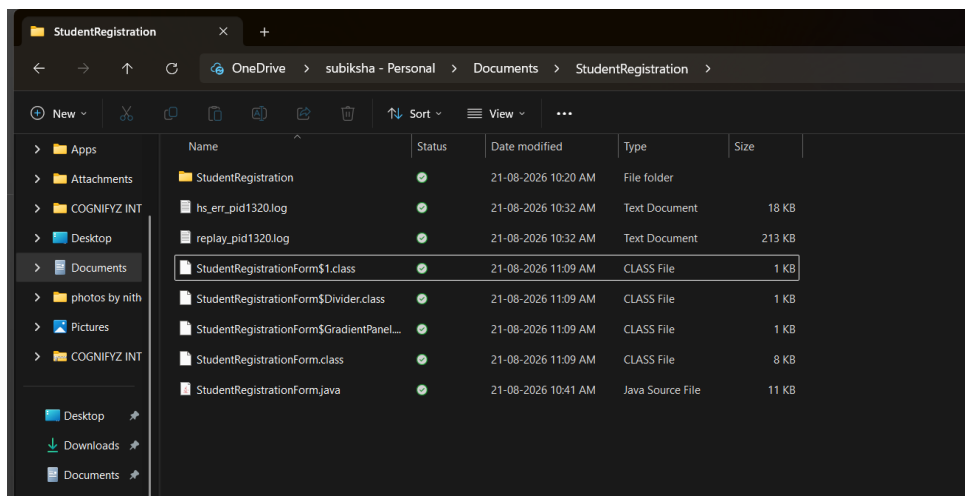
public static void main(String[] args) {

    new StudentRegistrationForm();

}

}

```



Student Registration | Elite Academy

**ELITE ACADEMY**  
STUDENT REGISTRATION

Full Name: Rose

Roll Number: 192424

Email Address: Roseli@gmail.com

Phone Number: 9829125267

Department: Computer Science

Course: B.E / B.Tech

Gender: ☐ Male ☒ Female

SUBMIT CLEAR EXIT

Registration Confirmed  
Name: Rose | Roll No: 192424 | Gender: Female  
Email: Roseli@gmail.com | Phone: 9829125267  
Department: Computer Science | Course: B.E / B.Tech

© Elite Academy | Office of Admissions

## 7. BorderLayout with Event Handling

### CODE :

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class BorderLayoutDemo extends Frame implements ActionListener
```

```
{
```

```
    Label header;
```

```
    TextArea display;
```

```
    Button home, about, exit;
```

```
    Button option1, option2;
```

```
    Button menu1, menu2;
```

```
    BorderLayoutDemo()
```

```
{
```

```
    setTitle("BorderLayout with Event Handling");
```

```
    setSize(600, 400);
```

```
    setLayout(new BorderLayout());
```

```
    header = new Label("STUDENT APPLICATION", Label.CENTER);
```

```
    add(header, BorderLayout.NORTH);
```

```
    display = new TextArea();
```

```
    display.setText("Welcome to the Student Application");
```

```
    add(display, BorderLayout.CENTER);
```



```
Panel southPanel = new Panel();

home = new Button("Home");
about = new Button("About");
exit = new Button("Exit");

southPanel.add(home);
southPanel.add(about);
southPanel.add(exit);

add(southPanel, BorderLayout.SOUTH);
Panel eastPanel = new Panel();
eastPanel.setLayout(new GridLayout(3, 1));

option1 = new Button("Option 1");
option2 = new Button("Option 2");

eastPanel.add(new Label("OPTIONS"));
eastPanel.add(option1);
eastPanel.add(option2);

add(eastPanel, BorderLayout.EAST);
Panel westPanel = new Panel();
westPanel.setLayout(new GridLayout(3, 1));

menu1 = new Button("Menu 1");
menu2 = new Button("Menu 2");

westPanel.add(new Label("MENU"));
westPanel.add(menu1);
westPanel.add(menu2);

add(westPanel, BorderLayout.WEST);
home.addActionListener(this);
about.addActionListener(this);
exit.addActionListener(this);
option1.addActionListener(this);
option2.addActionListener(this);
menu1.addActionListener(this);
menu2.addActionListener(this);

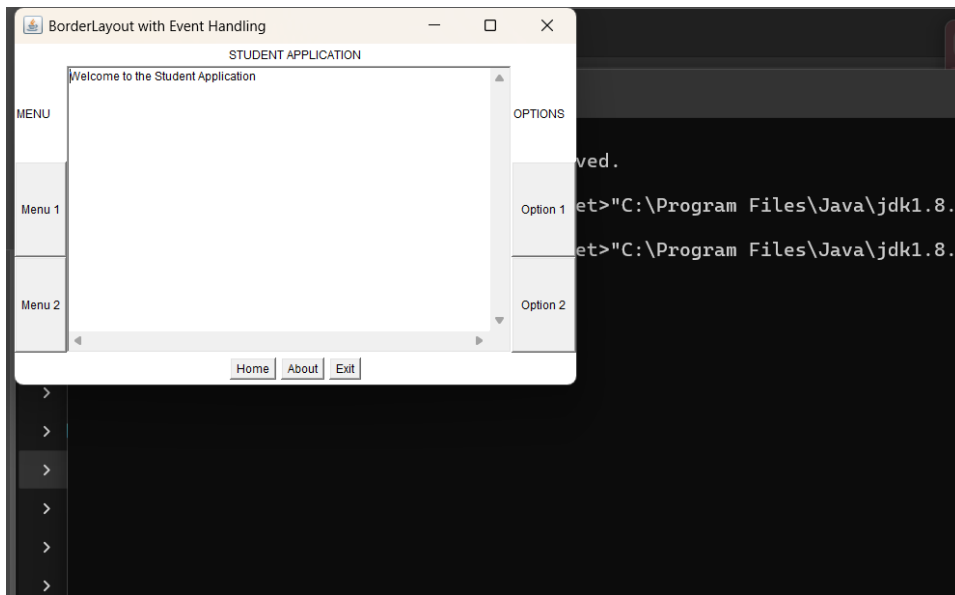
setVisible(true);
}
```

```

public void actionPerformed(ActionEvent e)
{
    if(e.getSource() == home)
    {
        display.setText("Home button clicked");
    }
    else if(e.getSource() == about)
    {
        display.setText("About button clicked");
    }
    else if(e.getSource() == option1)
    {
        display.setText("Option 1 selected");
    }
    else if(e.getSource() == option2)
    {
        display.setText("Option 2 selected");
    }
    else if(e.getSource() == menu1)
    {
        display.setText("Menu 1 selected");
    }
    else if(e.getSource() == menu2)
    {
        display.setText("Menu 2 selected");
    }
    else if(e.getSource() == exit)
    {
        System.exit(0);
    }
}

public static void main(String args[])
{
    new BorderLayoutDemo();
}
}

```



## 8. CardLayout with Event Handling

```
import java.awt.*;
import java.awt.event.*;
```

```
public class CardLayoutDemo extends Frame implements ActionListener
{
```

```
    CardLayout cardLayout;
    Panel cardPanel;
```

```
    TextField nameField;
    TextField regField;
```

```
    Choice department;
    Choice year;
    TextArea report;
    Button nextButton;
    Button previousButton;
    Button submitButton;
    public CardLayoutDemo() {
```

```
        setTitle("Student Registration - CardLayout");
        setSize(600, 450);
```

```
cardLayout = new CardLayout();
```

```
cardPanel = new Panel(cardLayout);
```

```
Panel card1 = new Panel(  
    new GridLayout(4, 2, 10, 10)  
);
```

```
card1.add(new Label("Student Information"));  
card1.add(new Label(""));
```

```
card1.add(new Label("Name:"));
```

```
nameField = new TextField();  
card1.add(nameField);
```

```
card1.add(new Label("Register Number:"));
```

```
regField = new TextField();  
card1.add(regField);
```

```
card1.add(new Label("Department:"));
```

```
department = new Choice();
```

```
department.add("AI & DS");  
department.add("CSE");  
department.add("ECE");  
department.add("IT");
```

```
card1.add(department);
```

```
Panel card2 = new Panel(  
    new GridLayout(3, 2, 10, 10)  
);
```

```
card2.add(new Label("Academic Information"));
card2.add(new Label(""));
```

```
card2.add(new Label("Year:"));
```

```
year = new Choice();
```

```
year.add("1st Year");
year.add("2nd Year");
year.add("3rd Year");
year.add("4th Year");
```

```
card2.add(year);
```

```
card2.add(new Label("Status:"));
card2.add(new Label("Active Student"));
```

```
Panel card3 = new Panel(
    new BorderLayout()
);
```

```
card3.add(
    new Label(
        "Confirmation / Report",
        Label.CENTER
    ),
    BorderLayout.NORTH
);
```

```
report = new TextArea();
```

```
card3.add(report, BorderLayout.CENTER);
cardPanel.add(card1, "Student");
cardPanel.add(card2, "Academic");
cardPanel.add(card3, "Confirmation");
```

```
add(cardPanel, BorderLayout.CENTER);
```

```

Panel buttonPanel = new Panel();

previousButton = new Button("Previous");
nextButton = new Button("Next");
submitButton = new Button("Submit");

previousButton.addActionListener(this);
nextButton.addActionListener(this);
submitButton.addActionListener(this);

buttonPanel.add(previousButton);
buttonPanel.add(nextButton);
buttonPanel.add(submitButton);

add(buttonPanel, BorderLayout.SOUTH);

addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        dispose();
    }
});

setVisible(true);
}
public void actionPerformed(ActionEvent e) {

    if (e.getSource() == nextButton) {

        cardLayout.next(cardPanel);
    }
    else if (e.getSource() == previousButton) {
        cardLayout.previous(cardPanel);
    }

    else if (e.getSource() == submitButton) {
        String studentName = nameField.getText();
        String registerNumber = regField.getText();

```

```

String dept = department.getSelectedItem();
String selectedYear = year.getSelectedItem();
report.setText(
    "STUDENT CONFIRMATION REPORT\n\n" +
    "Name: " + studentName + "\n" +
    "Register Number: " + registerNumber + "\n" +
    "Department: " + dept + "\n" +
    "Year: " + selectedYear + "\n\n" +
    "Registration Successful!"
);
cardLayout.show(
    cardPanel,
    "Confirmation"
);
}
}
public static void main(String[] args) {
    new CardLayoutDemo();
}
}

```

